EECS 16A Designing Information Devices and Systems I
Homework 13

Submission Format
Your homework submission should consist of **one** file.

- `hw13.pdf`: A single PDF file that contains all of your answers (any handwritten answers should be scanned) as well as your IPython notebook saved as a PDF.

  If you do not attach a PDF "printout" of your IPython notebook, you will not receive credit for problems that involve coding. Make sure that your results and your plots are visible. Assign the IPython printout to the correct problem(s) on Gradescope.

Submit the file to the appropriate assignment on Gradescope.

1. Audio File Matching

(Contributors: Aviral Pandey, Michael Kellman, Moses Won, Spencer Kent, Titan Yuan, Urmita Sikder, Vijay Govindarajan)

Learning Goal: *This problem motivates the application of correlation for pattern matching applications such as Shazam.*

Many audio processing applications rely on representing audio files as vectors, referred to as audio *signals*. Every component of the vector determines the sound we hear at a given time. We can use inner products to determine if a particular audio clip is part of a longer song, similar to an application like *Shazam*.

Let us consider a very simplified model for an audio signal, $\vec{x}$. At each timestep k, the audio signal can be either $x[k] = -1$ or $x[k] = 1$.

(a) Say we want to compare two audio files of the same length N to decide how similar they are. First, consider two vectors that are exactly identical, namely $\vec{x}_1 = \begin{bmatrix} 1 & 1 & \cdots & 1 \end{bmatrix}^T$ and $\vec{x}_2 = \begin{bmatrix} 1 & 1 & \cdots & 1 \end{bmatrix}^T$. What is the inner product of these two vectors? What if $\vec{x}_1 = \begin{bmatrix} 1 & 1 & \cdots & 1 \end{bmatrix}^T$ but $\vec{x}_2$ oscillates between 1 and -1? Assume that N, the length of the two vectors, is an even number.
Use this to suggest a method for comparing the similarity between a generic pair of length-N vectors.

(b) Next, suppose we want to find a short audio clip in a longer one. We might want to do this for an application like *Shazam*, which is able to identify a song from a short clip. Consider the vector of length 8, $\vec{x} = \begin{bmatrix} -1 & 1 & 1 & -1 & 1 & 1 & -1 & 1 \end{bmatrix}^T$.
We want to find the short segment $\vec{y} := \begin{bmatrix} y[0] & y[1] & y[2] \end{bmatrix}^T = \begin{bmatrix} 1 & 1 & -1 \end{bmatrix}^T$ in the longer vector. To do this, perform the linear cross correlation between these two finite length sequences and identify at what shift(s) the linear cross correlation is maximized. Apply the same technique to identify what shift(s) gives the best match for $\vec{y} = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^T$.
(If you wish, you may use iPython to do this part of the question, but you do not have to.)

(c) Now suppose our audio vector is represented using integers beyond simply just 1 and -1. Find the short audio clip $\vec{y} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}^T$ in the song given by $\vec{x} = \begin{bmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 & 10 \end{bmatrix}^T$. Where do you expect to see the peak in the correlation of the two signals? Is the peak where you want it to be, i.e. does it pull out the clip of the song that you intended? Why?
(If you wish, you may use iPython to do this part of the question, but you do not have to.)

(d) Let us think about how to get around the issue in the previous part. We applied cross-correlation to compare segments of $\vec{x}$ of length 3 (which is the length of $\vec{y}$) with $\vec{y}$. Instead of directly taking the cross correlation, we want to normalize each inner product computed at each shift by the magnitudes of both segments, i.e. we want to consider $\frac{<\vec{x}_k,\vec{y}>}{\|\vec{x}_k\|\|\vec{y}\|}$, where $\vec{x}_k$ is the length 3 segment starting from the k-th index of $\vec{x}$. This is referred to as normalized cross correlation. Using this procedure, now which segment matches the short audio clip best?

(e) We can use this on a more 'realistic' audio signal – refer to the IPython notebook, where we use normalized cross-correlation on a real song. Run the cells to listen to the song we are searching through, and add a simple comparison function `vector_compare` to find where in the song the clip comes from. Running this may take a couple minutes on your machine, but note that this computation can be highly optimized and run super fast in the real world! Also note that this is not exactly how Shazam works, but it draws heavily on some of these basic ideas.

2. Constrained Least Squares Optimization

In this problem, we will guide you through solving the following optimization problem:

Consider a matrix $\mathbf{A} \in \mathbb{R}^{M\times N}$ where $M > N$ and all N columns are linearly independent. Determine a unit vector $\hat{\vec{x}}$ that minimizes $\|\mathbf{A}\vec{x}\|$, where $\|\cdot\|$ denotes the norm—that is,

$$\|\mathbf{A}\vec{x}\|^2 \triangleq \langle \mathbf{A}\vec{x}, \mathbf{A}\vec{x}\rangle = (\mathbf{A}\vec{x})^T\mathbf{A}\vec{x} = \vec{x}^T\mathbf{A}^T\mathbf{A}\vec{x}.$$

This is equivalent to solving the following optimization problem:

$$\min_{\vec{x}} \|\mathbf{A}\vec{x}\|^2 \quad \text{subject to the constraint} \quad \|\vec{x}\|^2 = 1.$$

This task may *seem* like solving a standard least squares problem $\mathbf{A}\vec{x} = \vec{b}$ where $\vec{b} = \vec{0}$, but it is different. As an example, notice $\vec{x} = \vec{0}$ is *not* a valid solution to our problem because the norm of the zero vector does not equal one. Our optimization problem is a least squares problem with a constraint—hence the term *constrained least squares optimization*. The constraint can be visualized as limiting the vector $\vec{x}$ to lie on a unit circle (radius of the circle is one) if $N = 2$ and on a unit sphere if $N = 3$.

Let $(\lambda_1, \vec{v}_1), \ldots, (\lambda_N, \vec{v}_N)$ denote the eigenpairs (i.e., eigenvalue/eigenvector pairs) of $\mathbf{A}^T\mathbf{A}$. Assume that the eigenvalues are all real, distinct and indexed in an descending fashion—that is,

$$\lambda_1 > \cdots > \lambda_N.$$

Assume, too, that each eigenvector has been normalized to have unit length—that is, $\|\vec{v}_k\| = 1$ for all $k \in \{1, \ldots, N\}$.

(a) Show that $\lambda_N > 0$, i.e. all the eigenvalues are strictly positive.
Hint: Consider $\|\mathbf{A}\vec{v}\|^2$, where $\vec{v}$ is an eigenvector of $\mathbf{A}^T\mathbf{A}$ with eigenvalue λ.

(b) Consider two eigenpairs $(\lambda_k, \vec{v}_k)$ and $(\lambda_\ell, \vec{v}_\ell)$ corresponding to distinct eigenvalues of $\mathbf{A}^T\mathbf{A}$—that is, $\lambda_k \neq \lambda_\ell$. Prove that the corresponding eigenvectors $\vec{v}_k$ and $\vec{v}_\ell$ are orthogonal: $\vec{v}_k \perp \vec{v}_\ell$.
To help you get started, consider the two equations

$$\mathbf{A}^T\mathbf{A}\,\vec{v}_k = \lambda_k\,\vec{v}_k \tag{1}$$

and

$$\vec{v}_\ell^T\mathbf{A}^T\mathbf{A} = \lambda_\ell\,\vec{v}_\ell^T. \tag{2}$$

The second equation can be derived by taking the transpose of both sides in the eigenvalue equation $\mathbf{A}^T\mathbf{A}\vec{v}_\ell = \lambda_\ell \vec{v}_\ell$. Premultiply Equation 1 with $\vec{v}_\ell^T$, postmultiply Equation 2 with $\vec{v}_k$, compare the two, and explain how one may then infer that $\vec{v}_k$ and $\vec{v}_\ell$ are orthogonal, i.e. $\langle \vec{v}_k, \vec{v}_\ell \rangle = 0$.

Premultiplication by a vector $\vec{u}$ means multiplying an expression by $\vec{u}$ on the left. For example, premultiplying the matrix $\mathbf{A}$ by $\vec{u}$ gives $\vec{u}\mathbf{A}$. Postmultiplication means multiplying on the right. Remember that in general matrix-vector multiplication is not commutative, so those operations are not identical.

(c) The results of part (b) imply that the N eigenvectors of $\mathbf{A}^T\mathbf{A}$ are mutually orthogonal. A basis formed by vectors that are both (1) mutually orthogonal and (2) have unit length is called an orthonormal basis. Since the eigenvalues of $\mathbf{A}^T\mathbf{A}$ are distinct, and the corresponding eigenvectors are scaled to have a norm of one, the scaled eigenvectors form an orthonormal basis in $\mathbb{R}^N$. This means that we can express an arbitrary vector $\vec{x} \in \mathbb{R}^N$ as a linear combination of the eigenvectors $\vec{v}_1, \ldots, \vec{v}_N$, as follows:

$$\vec{x} = \sum_{n=1}^{N} \alpha_n \vec{v}_n.$$

i. Determine the n^{th} coefficient α_n in terms of $\vec{x}$ and one or more of the eigenvectors $\vec{v}_1, \ldots, \vec{v}_N$.

ii. Suppose $\vec{x}$ is a unit-length vector (i.e., a unit vector) in $\mathbb{R}^N$. Show that

$$\sum_{n=1}^{N} \alpha_n^2 = 1$$

where the α_n's are the coefficients of $\vec{x}$ in the basis defined earlier.
Hint: We know that $\|\vec{x}\|^2 = 1$. *Use the previous part to expand this expression.*

(d) Now express $\|\mathbf{A}\vec{x}\|^2$ in terms of $\{\alpha_1, \alpha_2 \ldots \alpha_N\}$, $\{\lambda_1, \lambda_2 \ldots \lambda_N\}$, and $\{\vec{v}_1, \vec{v}_2 \ldots \vec{v}_N\}$, and find an expression for $\vec{\hat{x}}$ such that $\|\mathbf{A}\vec{x}\|^2$ is minimized. Do *not* use any tool from calculus to solve this problem, so avoid differentiation.

Hint 1: Start with expressing $\mathbf{A}^T\mathbf{A}\vec{x}$ *in terms of* $\{\alpha_1, \alpha_2 \ldots \alpha_N\}$, $\{\lambda_1, \lambda_2 \ldots \lambda_N\}$, *and* $\{\vec{v}_1, \vec{v}_2 \ldots \vec{v}_N\}$, *then extend this to* $\|\mathbf{A}\vec{x}\|^2$.

Hint 2: After expressing $\|\mathbf{A}\vec{x}\|^2$ *in terms of* $\{\alpha_1, \alpha_2 \ldots \alpha_N\}$, $\{\lambda_1, \lambda_2 \ldots \lambda_N\}$, *and* $\{\vec{v}_1, \vec{v}_2 \ldots \vec{v}_N\}$, *which variable are you minimizing over?*

3. Noise Canceling Headphones

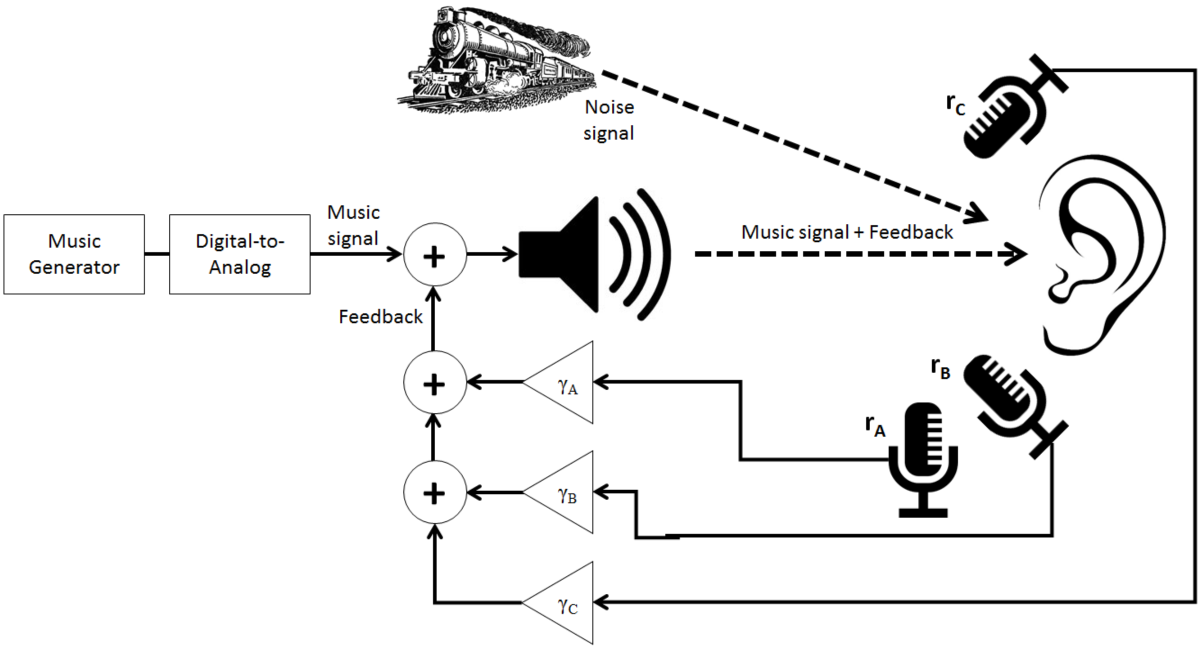
(Contributors: Aviral Pandey, Lydia Lee, Michael Kellman, Moses Won, Urmita Sikder, Wahid Rahman)

Learning Goal: *This problem will employ least squares method to minimize the effect of additive noise through headphones.*

In this problem, we will explore a common design for noise cancellation using noise-canceling headphones as an example application. We will work with the model shown in the figure below.

A music signal is generated at a speaker and transmitted to the listener's ear. If there is noise in the environment (*e.g.* other people's voices, a train going by), this noise signal will be superimposed (i.e. added) on the music signal and the listener will hear both. In order to cancel the noise, we will try **to record the noise and subtract it directly from the transmitted signal** with the hope that we can achieve perfect cancellation of everything but the music. Since our system is imperfect, we'll have to solve a **least squares problem**.

The gain blocks marked by γ **(Greek "gamma") represent scalar multiplication**, and we will assume that they can take on any real number, positive or negative.



Assume we have an **additive noise signal** noted by $\vec{n}$, which we want to cancel. This represents the extra noise coming from the train.

$$\vec{n} = \begin{bmatrix} n_1 \\ n_2 \\ n_3 \\ n_4 \\ n_5 \end{bmatrix}.$$

Let us assume the music signal is represented by

$$\vec{m} = \begin{bmatrix} m_1 \\ m_2 \\ m_3 \\ m_4 \\ m_5 \end{bmatrix}.$$

In absence of any noise cancellation, the user will hear:

$$\vec{s} = \vec{m} + \vec{n}.$$

However, we want to cancel this noise. For this we use three microphones to record this noise, Mic A, Mic B, and Mic C. However, they cannot perfectly record the noise $\vec{n}$ and have erroneous recordings. Let $\vec{r}_A$, $\vec{r}_B$, and $\vec{r}_C$ represent the **noise that each microphone picks up**:

$$\vec{r}_A = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{bmatrix}, \vec{r}_B = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \end{bmatrix}, \vec{r}_C = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \end{bmatrix}.$$

We can arrange the recordings into a matrix $\mathbf{R}$ and the microphone gains, γ, into a vector $\vec{\gamma}$ to get:

$$\mathbf{R} = \begin{bmatrix} \vec{r}_A & \vec{r}_B & \vec{r}_C \end{bmatrix} = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \\ a_4 & b_4 & c_4 \\ a_5 & b_5 & c_5 \end{bmatrix}, \vec{\gamma} = \begin{bmatrix} \gamma_A \\ \gamma_B \\ \gamma_C \end{bmatrix}.$$

We want to choose a $\vec{\gamma}$ such that we minimize the impact of the noise.

The listener's ear will hear a total signal of:

$$\vec{s} = \vec{m} + \mathbf{R}\vec{\gamma} + \vec{n},$$

$$\begin{bmatrix} s_1 \\ s_2 \\ s_3 \\ s_4 \\ s_5 \end{bmatrix} = \begin{bmatrix} m_1 \\ m_2 \\ m_3 \\ m_4 \\ m_5 \end{bmatrix} + \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \\ a_4 & b_4 & c_4 \\ a_5 & b_5 & c_5 \end{bmatrix} \begin{bmatrix} \gamma_A \\ \gamma_B \\ \gamma_C \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \\ n_3 \\ n_4 \\ n_5 \end{bmatrix}.$$

This problem focuses on how to choose $\vec{\gamma}$ so as to have the signal $\vec{s}$ be as close to $\vec{m}$ as possible.

(a) Ideally, we would want to have a signal at the ear that matches the original music signal $\vec{m}$ perfectly. In reality, this is not possible, so we will aim to minimize the effect of the noise. What quantity would we need to minimize to make sure this happens? Write your answer in terms of the matrix $\mathbf{R}$, the vector of mic gains $\vec{\gamma}$, and the noise vector $\vec{n}$. *Hint: Your answer should be the norm of something.*

(b) We can solve the noise minimization problem by the least squares method. In effect, if we have a problem, $\min_{\vec{x}} \left\| \mathbf{A}\vec{x} - \vec{b} \right\|$, then the $\vec{x}$ that solves this problem is,

$$\vec{\hat{x}} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\vec{b}. \tag{3}$$

Implement this least squares solution in the IPython Notebook helper function `doLeastSquares`.
Remember that $\min_{\vec{x}} \left\| \mathbf{A}\vec{x} + \vec{b} \right\| = \min_{\vec{x}} \left\| \mathbf{A}\vec{x} - (-\vec{b}) \right\|$.

(c) For the given $\vec{n}$ and the recordings, $\vec{r}_A$, $\vec{r}_B$, $\vec{r}_C$, below, find the γ's that minimize the effect of noise. Use the helper function `doLeastSquares` you made in the last part to solve this.

$$\vec{n} = \begin{bmatrix} 0.10 \\ 0.37 \\ -0.45 \\ 0.068 \\ 0.036 \end{bmatrix}, \vec{r}_A = \begin{bmatrix} 0 \\ 0.11 \\ -0.31 \\ -0.012 \\ -0.018 \end{bmatrix}, \vec{r}_B = \begin{bmatrix} 0 \\ 0.22 \\ -0.20 \\ 0.080 \\ 0.056 \end{bmatrix}, \vec{r}_C = \begin{bmatrix} 0 \\ 0.37 \\ -0.44 \\ 0.065 \\ 0.038 \end{bmatrix}.$$

The next few questions can be answered in the IPython notebook by running the associated cells.

(d) Follow the instructions in the IPython notebook to load a **music signal** and some **noise signals**. Listen to the music signal `music_y` and the two noise signals `noise1_y` and `noise2_y`. Which ones are full of static and which ones are not?

Also listen to the **noisy music signal** `noisyMusic`, generated by adding the music signal `music_y` and the first noise signal `noise1_y`. What do you hear?

(e) For this part we assume the noise signal $\vec{n}$ is given by `noise1_y` only. **R**, i.e. the matrix of recorded noise from 3 microphones, is simulated with the `recordAmbientNoise` function. Calculate a vector $\vec{\gamma}$ that minimizes the effect of noise using `doLeastSquares`.

(f) Use you answers from the previous part to find the **noise-cancellation signal $\mathbf{R}\vec{\gamma}$**. Add this noise-cancellation signal ($\mathbf{R}\vec{\gamma}$), the music signal $\vec{m}$ (`music_y`), and the noise signal $\vec{n}$ (`noise1_y`), in order to generate a **noise-cancelled signal**. Play the noisy signal from part (e) and the noise-cancelled signal. Can you hear a difference?

(g) **[Optional]** Try adding the other noise signal `noise2_y` to the music signal to generate a new noisy signal. Don't re-calculate new values for $\vec{\gamma}$ i.e. don't solve the least squares problem again. Recalculate **R** and find the new noise-cancellation signal $\mathbf{R}\vec{\gamma}$ using $\vec{\gamma}$ from part (f). Add this noise-cancellation signal to your noisy signal. Comment on the quality of the resulting noise-cancelled signal. Is it perfect or are there artifacts?

Note: We want to find out how the $\vec{\gamma}$ we calculated work for any type of noise.

4. Course Evaluations [Optional]

Please take a moment to fill out course evaluations if you have not done so already! It really helps us improve the course for future semesters! Links to this should have already been sent to your email!